

DIGITAL TWIN FOR PERSONALIZED HEALTHCARE

PHASE-WISE PROJECT EXECUTION ROADMAP

AI • Healthcare • Digital Twin • Full-Stack Development • Research

Purpose: Build the project incrementally from research and MVP to an AI-powered, interactive Digital Twin prototype and research paper.

Phase 0 — Research & Planning

Timeline: Week 1

Goal: Finalize problem statement, objectives, literature review, research gap, users, scope, architecture, datasets and backlog.

Deliverables: Requirements document; literature matrix; initial architecture; GitHub repository.

Phase 1 — UI/UX & Product Design

Timeline: Weeks 1-2

Goal: Design landing page, authentication, onboarding, dashboard, Digital Twin, reports, AI assistant, doctor/admin portals and responsive layouts.

Deliverables: Wireframes; design system; clickable prototype.

Suggested Technologies: Next.js, React, Tailwind, Figma

Phase 2 — Foundation & Authentication

Timeline: Week 2

Goal: Set up frontend/backend, database, environment variables, registration, login, sessions/JWT, RBAC and protected routes.

Deliverables: Working patient/doctor/researcher/admin authentication.

Suggested Technologies: Next.js, FastAPI, PostgreSQL, JWT

Phase 3 — Patient Profile & Health Data

Timeline: Week 3

Goal: Build personal profile, medical/family history, allergies, medications, vaccinations, lifestyle, vitals and health goals.

Deliverables: Complete patient data layer.

Suggested Technologies: React, FastAPI, PostgreSQL

Phase 4 — Medical Records & OCR

Timeline: Week 4

Goal: Secure PDF/image upload, OCR, medical field extraction, lab-value parsing, reference ranges, AI summarization and user verification.

Deliverables: Upload → extraction → verification → database pipeline.

Suggested Technologies: Python, OCR, LLM, object storage

Phase 5 — Dashboard & Timeline

Timeline: Week 5

Goal: Build health overview, charts, lab/vital trends, medication timeline, recent reports, filters and AI insights.

Deliverables: Functional patient dashboard.

Suggested Technologies: Next.js, chart library, PostgreSQL

Phase 6 — AI Prediction Engine

Timeline: Weeks 6-7

Goal: Choose 1-2 diseases, preprocess data, engineer features, train baseline + tree models, evaluate, serialize and expose prediction APIs.

Deliverables: Working disease-risk model + evaluation report.

Suggested Technologies: Python, Scikit-learn, XGBoost, FastAPI

Phase 7 — Explainable AI & Health Score

Timeline: Week 7

Goal: Add SHAP/LIME or feature importance, risk-factor explanations and a transparent prototype wellness score.

Deliverables: Explainable prediction UI and score breakdown.

Suggested Technologies: SHAP/LIME, Python, React

Phase 8 — AI Assistant & RAG

Timeline: Week 8

Goal: Parse documents, create embeddings, retrieve relevant patient context, generate grounded LLM answers, maintain conversations and add safety guardrails.

Deliverables: RAG assistant with document/source references.

Suggested Technologies: LLM, LangChain, vector DB, FastAPI

Phase 9 — 3D Digital Twin

Timeline: Week 9

Goal: Create interactive human model, organ selection, organ data panels, health overlays, historical state and performance-optimized animations.

Deliverables: Interactive Digital Twin connected to patient data.

Suggested Technologies: Three.js, React Three Fiber, GLTF/GLB

Phase 10 — Future Health Simulator

Timeline: Week 10

Goal: Create what-if scenarios for modifiable factors such as weight, activity and sleep; compare baseline and hypothetical trends with assumptions.

Deliverables: Prototype scenario simulator with clear hypothetical-output labeling.

Suggested Technologies: ML, FastAPI, React charts

Phase 11 — Doctor Portal

Timeline: Weeks 10–11

Goal: Patient search, overview, reports, timeline, Digital Twin, AI predictions/explanations, doctor notes and export.

Deliverables: Functional clinician review workflow.

Suggested Technologies: Next.js, FastAPI, RBAC

Phase 12 — Advanced Integrations

Timeline: Weeks 11–12

Goal: Add simulated wearable stream, selected wearable APIs, medical-image AI, voice, multilingual support, reminders and notifications.

Deliverables: Selected advanced features without weakening MVP.

Suggested Technologies: Web APIs, vision/speech models

Phase 13 — Security, Testing & Quality

Timeline: Week 12

Goal: Unit/API/UI testing, AI validation, permission tests, file security, rate limits, audit logs, privacy controls and performance tests.

Deliverables: Test report, bug tracker and release candidate.

Suggested Technologies: Pytest, Playwright, monitoring

Phase 14 — Deployment & DevOps

Timeline: Weeks 12–13

Goal: Dockerize services, configure secrets, CI/CD, frontend/backend/database/storage deployment, HTTPS, monitoring and backups.

Deliverables: Live demo + deployment documentation.

Suggested Technologies: Docker, GitHub Actions, cloud hosting

Phase 15 — Research & Paper

Timeline: Weeks 13–15

Goal: Run model comparisons, ablation studies, explainability/usability evaluation, analyze results, limitations and future work.

Deliverables: Experiments, figures, tables and IEEE-style paper draft.

Suggested Technologies: Python, Jupyter, statistics

Phase 16 — Final Release

Timeline: Weeks 15–16

Goal: Prepare demo, PPT, poster, GitHub README, final report, viva questions, backup demo and project video.

Deliverables: Final deployed prototype + documentation + presentation.

Suggested Technologies: Documentation/presentation tools

Priority Levels

Level	Features	Purpose
P0 — MVP	Auth, profile, records, OCR, dashboard, 1-2 ML models, AI assistant	Complete basic Digital Twin product
P1 — Differentiator	Explainable AI, 3D Twin, health score, doctor portal, simulator	Make the project stand out
P2 — Advanced	Wearables, medical imaging, voice, multilingual, FHIR	Add research/industry depth
P3 — Future	Federated learning, multimodal models, hospital integrations	Long-term research direction

Critical Rule

Build the data foundation before the 3D experience. Recommended order: authentication → database → patient data → medical records/OCR → APIs → ML → RAG → Digital Twin → advanced features → testing/deployment → research.

MVP Finish Line

Login → onboarding → upload report → OCR + summary → dashboard → timeline → disease-risk prediction → explanation → AI assistant → basic 3D Digital Twin → PDF report.